

Sample End Semester Question Paper - 2025

Subject: Linear Algebra and Probability | Code: AIML201 | Semester: 2
Branch: AI & ML | Regulation: R20 (sample) | Time: 3 Hours | Max Mar

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What is a basis?
2. Define linear independence.
3. What is a stochastic matrix?
4. State the law of total probability.
5. Define a random variable.
6. Find the rank of a 3x3 matrix (given).
7. State Cramer's rule.
8. What is an eigenvalue?
9. Define a vector space.
10. State one property of determinants.

PART B - Long questions (6 x 13 = 78 Marks)

1. Solve a system of linear equations by Gauss elimination.
2. Find eigenvalues and eigenvectors of a given matrix. Check diagonalisability.
3. Explain determinants and Cramer's rule with examples.
4. Explain linear independence and basis of a vector space.
5. Solve a least squares problem with an example.
6. Explain basic probability: addition, multiplication and conditional probability with examples.